

Generation Interconnection System Impact Study Report

For Queue Project Q21-1861

Colton 138 kV
August 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1861. The Interconnection Customer proposes a 155.0 MW wind facility interconnecting at the Colton 138 kV substation in the West Virginia area. The request was received on August 17, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 650 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,800,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Mcclain Energy Facility - De Kalb 138 kV: 136.0% loading (220 MVA) following the outage of the Central Juncos 3202 - Colton 138 kV L; DFAX = 51.9%.

The De Kalb - Oxnard 138 kV is overloaded to 131.3% of its 223 MVA emergency rating for the loss of the Colton - Central Juncos 3202 138 kV. The project contributes 51.9% DFAX (80.5 MW) to this overload.

Hinton Ap - Oxnard 138 kV: 124.4% loading (229 MVA) following the outage of the Colton - Central Juncos 3202 138 kV; DFAX = 51.9%.

Kimper - Tom Watkins 138 kV: 119.4% loading (396 MVA) following the outage of the Atoka - Catawba 345 kV; DFAX = 6.9%.

The Haysi - Leesburg Picciola 138 kV Line is overloaded to 113.9% of its 269 MVA emergency rating for the loss of the Atoka - Catawba 345 kV. The project contributes 5.6% DFAX (8.6 MW) to this overload.

For the loss of Atoka Substation - Catawba 345 kV, the Tom Watkins - Haysi 138 kV L loads to 113.6% of its 349 MVA rating. Project DFAX: 9.6%.

The Leavitt - Chancellor Row 345 kV is overloaded to 102.8% of its 2,057 MVA emergency rating for the loss of the Leavitt - Birchwood 345 kV. The project contributes 5.1% DFAX (7.8 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcclain Energy Facility - De Kalb 138 kV	Rebuild line with higher-capacity conductor	\$3,300,000	18% (\$585,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
De Kalb - Oxnard 138 kV	Rebuild line with higher-capacity conductor	\$8,626,000	18% (\$1,528,000)
Hinton Ap - Oxnard 138 kV	Rebuild line with higher-capacity conductor	\$4,709,000	18% (\$834,000)
Tom Watkins - Kimper 138 kV	Rebuild line with higher-capacity conductor	\$9,472,000	2% (\$238,000)
Haysi - Leesburg Picciola 138 kV Line	Rebuild line with higher-capacity conductor	\$19,035,000	3% (\$567,000)
Tom Watkins - Haysi 138 kV L	Rebuild line with higher-capacity conductor	\$37,755,000	2% (\$915,000)
Leavitt - Chancellor Row 345 kV	Reconductor line	\$15,650,000	1% (\$133,000)
	Total allocated to project		\$4,800,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.